STA2002: Probability and Statistics II
Chi-square Goodness-of-fit Tests

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Introduction

- In this lecture we will learn the very important **chi-square test statistic** first proposed by Karl Pearson in 1900. Also would consider applications of this chi-square test statistic which can be used for many different types of tests. In particular, one application allows us to test the appropriateness of different probabilistic models.

- **Suggested reading**: Chapter 9.1 of the textbook.

Karl Pearson

- Karl Pearson (27 March 1857 – 27 April 1936) was an English mathematician and statistician, well known for, Principal component analysis, Pearson distribution, Pearson's chi-squared test, Phi coefficient, Chi-square distribution, Contingency table, Histogram, Kurtosis, Mode, Random walk, etc.



Introduction

- Let $Y_1 \sim Bin(n, p_1)$, and $Y_2 = n - Y_1 \sim Bin(n, p_2)$ with $p_2 = 1 - p_1$.
- For example, Y_1 can be the number of smokers in a survey of $n = 100$ people, and Y_2 is the number of non-smokers in the same survey, and there are only 2 **categories** in this case.
- When n is large, due to Central Limit Theorem, we have,

$$Y_1 \overset{approx}{\sim} N(np_1, np_1(1-p_1)),$$

and then,

$$Q_1 = \frac{(Y_1 - np_1)^2}{np_1(1-p_1)} \overset{approx}{\sim} \chi^2(1).$$

Introduction

- Meanwhile, Q_1 can be written as

$$
\begin{aligned}
Q_1 &= \frac{(Y_1 - np_1)^2}{np_1(1-p_1)} \sim \chi^2(1) \\
&= \frac{(Y_1 - np_1)^2}{np_1(1-p_1)}(1 - p_1 + p_1) \\
&= \frac{(Y_1 - np_1)^2}{np_1} + \frac{(Y_1 - np_1)^2}{n(1-p_1)} \\
&= \frac{(Y_1 - np_1)^2}{np_1} + \frac{(Y_2 - np_2)^2}{np_2} \\
&= \sum_{i=1}^{2} \frac{(Y_i - np_i)^2}{np_i} \overset{\text{approx}}{\sim} \chi^2(1).
\end{aligned}
$$

$\because Y_1 + Y_2 = n,\ p_1 + p_2 = 1$

$\therefore (Y_1 - np_1) = (n - Y_2 - n(1-p_2)) = (-Y_2 - np_2)$

$n(1-p_1) = np_2.$

$\frac{\frac{Y_1}{n} - p_1}{\sqrt{p_1(1-p_1)/n}} \sim N(0,1)$ by CLT.

Introduction

- Actually, we have already showed chi-square test statistic with only 2 categories.
- Next, we would generalize the form of the test statistic for k categories, with $k \geq 2$.
- Suppose that the population or the experiment can be split into k (mutually exclusive and exhaustive) **categories**, say $A_1, A_2, \ldots, A_k$, and Y_i is the number of times that is in category i for $i = 1, 2, \ldots, k$.
- Let $p_i = P(A_i)$, $i = 1, 2, \ldots, k$, with $\sum_{i=1}^{k} p_i = 1$.

Chi-square Test

- We would like to test,

$$H_0 : p_i = p_{i0}, \quad i = 1, 2, \ldots, k,$$

against

$$H_1 : \text{All alternative possibilities},$$

where p_{i0}, $i = 1, 2, \ldots, k$, denotes fixed constant.

- **Rule of thumbs**: Suppose that $np_i \geq 5$ for $i = 1, 2, \ldots, k$.
- Question: Consider what is a suitable test statistic, and the relative natural form of critical region?

Test statistic

- Construct the following test statistic,

$$Q_{k-1} = \sum_{i=1}^{k} \frac{(Y_i - np_{i0})^2}{np_{i0}}.$$

- Under H_0, $Q_{k-1} \overset{approx}{\sim} \chi^2(k-1)$.
- Once data is collected, calculate the realized test statistic,

$$q_{k-1} = \sum_{i=1}^{k} \frac{(y_i - np_{i0})^2}{np_{i0}} = \sum_{i=1}^{k} \frac{(\text{Observed}_i - \text{Expected}_i)^2}{\text{Expected}_i}.$$

↳ follow chī-square.

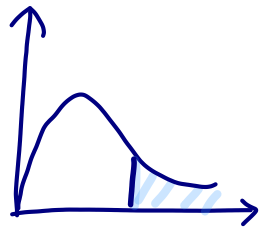
Decision Rule

- With significance level α, reject H_0 when,

$$q_{k-1} \geq \chi^2_\alpha(k-1).$$

- **Critical region**,

$$C := \{Q_{k-1} \geq \chi^2_\alpha(k-1)\}.$$



- **p-value**,

$$\text{p-value} = P(Q_{k-1} \geq q_{k-1};\ H_0).$$

Pay attention to the d.f.

- Pay attention, here the degree of freedom is $k-1$. There are only $k-1$ independent Y_is, as $Y_k = n - Y_1 - \cdots - Y_{k-1}$.
- Pay attention, here the degree of freedom is $k-1$. Or we would say, there are only $k-1$ independent and unknown parameters p_is, as $\sum_{i=1}^{k} p_i = 1$.
- Rigorous proofs of the asymptotic distribution of the chi-square test statistic is not required in this course.

Multimonial Distribution

- The joint pmf of $Y_1, \ldots, Y_k$ is, the marginal of $y_i \sim$ Binomial

$$f(y_1, y_2, \ldots, y_k) = \frac{n!}{y_1! \cdots y_k!} p_1^{y_1} p_2^{y_2} \cdots p_k^{y_k}$$

  with $\sum_{i=1}^{k} y_i = n$ and $\sum_{i=1}^{k} p_i = 1$.

- This is the pmf of a **Multinomial Distribution** with parameters $p_1, p_2, \ldots, p_{k-1}, p_k = 1 - \sum_{i=1}^{k-1} p_i$,
  $p_i \geq 0$, $i = 1, 2, \ldots, k$,
  and $y_i = 0, 1, \ldots, n$, $i = 1, 2, \ldots, k$, $\sum_{i=1}^{k} y_k = n$.

Example 9.1-2

Example 1

Let X denote the number of heads that occur when four coins are tossed at random.

Under the assumption that the four coins are independent and the probability of heads on each coin is $1/2$, then,

$$X \sim \text{Binomial}(4, 1/2).$$

Question: How to use data thus to test this hypothesis or support this assumption?

Example 9.1-2

- One hundred repetitions of this experiment resulted in 0, 1, 2, 3, and 4 heads being observed on 7, 18, 40, 31, and 4 trials, respectively.
- Question: Do these results support the assumptions?
- Or equivalently, is Binomial$(4, 1/2)$ a reasonable model for the distribution of X?

Example 9.1-2

Solution.

- Next we conduct the analysis, we begin by letting $A_1 = \{0\}$, $A_2 = \{1\}$, $A_3 = \{2\}$, $A_4 = \{3\}$, and $A_5 = \{4\}$.
- Under the assumption that $X \sim \text{Binomial}(4, 1/2)$,

$$p_{10} = p_{50} = \binom{4}{0}\left(\frac{1}{2}\right)^4 = \frac{1}{16} = 0.0625,$$

$$p_{20} = p_{40} = \binom{4}{1}\left(\frac{1}{2}\right)^4 = \frac{4}{16} = 0.25,$$

$$p_{30} = \binom{4}{2}\left(\frac{1}{2}\right)^4 = \frac{6}{16} = 0.375.$$

Example 9.1-2

- The null hypothesis is proposed as,

$$H_0 : p_i = p_{i0}, \quad i = 1, 2, \ldots, 5,$$

against

$$H_1 : \text{All alternative possibilities},$$

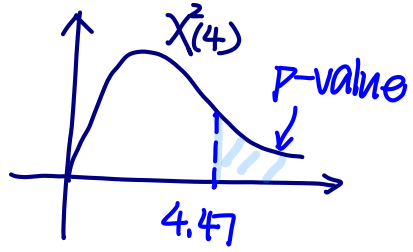
- Define test statistic Q_4, and the realized value is,

$$\begin{aligned} q_4 &= \frac{(7-6.25)^2}{6.25} + \frac{(18-25)^2}{25} + \frac{(40-37.5)^2}{37.5} + \frac{(31-25)^2}{25} + \frac{(4-6.25)^2}{6.25} \\ &= 4.47. \end{aligned}$$

Example 9.1-2

- At significance level $\alpha = 5\%$, $\chi^2_{0.05}(4) = 9.488$.
- As,

$$q_4 = 4.47 < \chi^2_{0.05}(4) = 9.488,$$



the null hypothesis is not rejected. if p-value $< \alpha$, reject H_0.!

- That is, the data support the hypothesis that Binomial$(4, 1/2)$ is a reasonable probabilistic model for X.
- Recall that the mean of a chi-square random variable is its number of degrees of freedom. In this example, the mean is 4 and the observed value of Q_4 is 4.47, just a little greater than the mean. □

Example 9.1-3

Example 2

Let X denote the number of alpha particles emitted by barium-133 in one tenth of a second.

The following 50 observations of X were taken with a Geiger counter in a fixed position:

7	4	3	6	4	4	5	3	5	3
5	5	3	2	5	4	3	3	7	6
6	4	3	11	9	6	7	4	5	4
7	3	2	8	6	7	4	1	9	8
4	8	9	3	9	7	7	9	3	10

The experimenter is interested in determining whether X has a Poisson distribution.

Example 9.1-3

- The null hypothesis is proposed as,

$$H_0: \ X \text{ is Poisson.}$$

- We then partition the set of outcomes for this experiment into the sets $A_1 = \{0, 1, 2, 3\}$, $A_2 = \{4\}$, $A_3 = \{5\}$, $A_4 = \{6\}$, $A_5 = \{7\}$, and $A_6 = \{8, 9, 10, \ldots\}$.
- Note that we combined $\{0, 1, 2, 3\}$ into one set A_1 and $\{8, 9, 10, \ldots\}$ into another A_6 so that the expected number of outcomes for each set would be at least five when H0 is true.

Example 9.1-3

- Recall that the mle of the parameter in Poisson(λ) is,

$$\hat{\lambda} = \overline{X}.$$

- Here, the realized sample mean is, $\overline{x} = 5.4$.
- Recall that the pmf of Poisson(λ) is,

$$f(y) = \frac{e^{-\lambda}\lambda^{y}}{y!}, \quad y = 0, 1, 2, \ldots,$$

Example 9.1-3

contigency table (multi-nomial is assumed)

	Outcome					
	A_1	A_2	A_3	A_4	A_5	A_6
Frequency	13	9	6	5	7	10
Probability	0.213	0.160	0.173	0.156	0.120	0.178
Expected ($50p_i$)	10.65	8.00	8.65	7.80	6.00	8.90

- Next calculate the realized value of $Q_{6-1} = Q_5$,

$$q_5 = \frac{[13 - 50(0.213)]^2}{50(0.213)} + \cdots + \frac{[10 - 50(0.178)]^2}{50(0.178)}$$

$$= 2.763 < 9.488 = \chi^2_{0.05}(4),$$

Example 9.1-3

- The decision is that: H_0 is not rejected at the 5% significance level.
- That is, with only these data, we are quite willing to accept the model that X has a Poisson distribution.
- Pay attention, here the d.f is $6 - 1 - 1 = 4$!
  As under H_0, p_i $i = 1, 2, \ldots, 6$ are functions of another parameter λ. Under $H_0 \cup H_1$, $\sum_{i=1}^{6} p_i = 1$, there are 5 independent and unknown parameters. While under H_0, there are only 1 independent and unknown parameter λ. The d.f. is $5 - 1 = 4$.

Example 9.1-4

Example 3

Given 105 observations of the times in minutes between calls to 911. We shall use a chi-square goodness-of-fit test to see whether or not the exponential pdf with $\theta = 20$ is an appropriate model for the data.

30	17	65	8	38	35	4	19	7	14	12	4	5	4	2
7	5	12	50	33	10	15	2	10	1	5	30	41	21	31
1	18	12	5	24	7	6	31	1	3	2	22	1	30	2
1	3	12	12	9	28	6	50	63	5	17	11	23	2	46
90	13	21	55	43	5	19	47	24	4	6	27	4	6	37
16	41	68	9	5	28	42	3	42	8	52	2	11	41	4
35	21	3	17	10	16	1	68	105	45	23	5	10	12	17

Example 9.1-4

- Solution.
  Recall the pdf of exponential distirubtion,

$$f(x;\theta) = \frac{1}{\theta} e^{-x/\theta}, \quad x \geq 0, \ \theta > 0,$$

  and the cdf is,

$$F(x;\theta) = 1 - e^{-x/\theta}, \quad x \geq 0, \ \theta > 0.$$

- We groups the data into nine classes, thus to define A_i, $i = 1, 2, \ldots, 9$. See the table attached.
- The realized value of the chi-square goodness-of-fit statistic is,

$$q_8 = \frac{(41 - 38.0520)^2}{38.0520} + \frac{(22 - 24.2655)^2}{24.2655} + \cdots + \frac{(2 - 2.8665)^2}{2.8665} = 4.6861.$$

Example 9.1-4

Table 9.1-2 Summary of times between calls to 911

Class	Frequency	Probability	Expected
$A_1 = [0, 9]$	41	0.3624	38.0520
$A_2 = (9, 18]$	22	0.2311	24.2655
$A_3 = (18, 27]$	11	0.1473	15.4665
$A_4 = (27, 36]$	10	0.0939	9.8595
$A_5 = (36, 45]$	9	0.0599	6.2895
$A_6 = (45, 54]$	5	0.0382	4.0110
$A_7 = (54, 63]$	2	0.0244	2.5620
$A_8 = (63, 72]$	3	0.0155	1.6275
$A_9 = (72, \infty)$	2	0.0273	2.8665

Example 9.1-4

- In this example, denote the cdf of the distribution assumed as F_0, then we have,

$$F_0(x) = 1 - e^{-x/20}, \quad x \geq 0.$$

- The value of p_{i0}, $i = 1, 2, \ldots, 9$, under H_0 can be prepared. For example,

$$\begin{aligned} p_{10} &= P(A_1; F_0) = F_0(9) = 0.3624, \\ p_{20} &= P(A_2; F_0) = F_0(18) - F_0(9) = 0.2311, \\ p_{30} &= P(A_3; F_0) = F_0(27) - F_0(18) = 0.1473, \end{aligned}$$

- See the table attached for other values of p_{i0}, $i = 1, 2, \ldots, 9$.

Example 9.1-4

- Compare the realized statistic with $\chi^2_\alpha(9-1) = \chi^2_\alpha(8)$.
- When $\alpha = 0.05$, $\chi^2_{0.05}(8) = 15.507$.
- Equivalently, we can calculate p-value, p-value$= 0.7905$.
- Decision: Retain H_0, which means that the exponential distribution with $\theta = 20$ is an pretty good fit.
- Pay attention: here the d.f. is 9 - 1 = 8. If under the H_0, parameter θ is unknown, then the d.f. is (9-1)-1 = 7.

□